Clustering I

Statistical Data Mining II

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Outline

- Similarity
- Dissimilarity
- Optimization Problem
- Combinatorial Algorithms

Introduction

- Very broad topic.
- Broad notion of Modularity.
- Ill-defined.
- Difficult to quantify and characterize.

Data Clustering: A Review

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Clustering is the unsupervised classification of patterns (observations, data items, or feature vectors) into groups (clusters). The clustering problem has been addressed in many contexts and by researchers in many disciplines; this reflects its broad appeal and usefulness as one of the steps in exploratory data analysis. However, clustering is a difficult problem combinatorially, and differences in assumptions and contexts in different communities has made the transfer of useful generic concepts and methodologies slow to occur. This paper presents an overview of pattern clustering methods from a statistical pattern recognition perspective, with a goal of providing useful advice and references to fundamental concepts accessible to the broad community of clustering practitioners. We present a taxonomy of clustering techniques, and identify cross-cutting themes and recent advances. We also describe some important applications of clustering algorithms such as image segmentation, object recognition, and information retrieval.

Categories and Subject Descriptors: I.5.1 [**Pattern Recognition**]: Models; I.5.3 [**Pattern Recognition**]: Clustering; I.5.4 [**Pattern Recognition**]: Applications—*Computer vision*; H.3.3 [**Information Storage and Retrieval**]: Information Search and Retrieval—*Clustering*; I.2.6 [**Artificial Intelligence**]: Learning—*Knowledge acquisition*

General Terms: Algorithms

Additional Key Words and Phrases: Cluster analysis, clustering applications, exploratory data analysis, incremental clustering, similarity indices, unsupervised learning

Introduction

Definition: The unsupervised classification of patterns (observations, data items, or feature vectors) into groups/modules/clusters. **OR** The grouping of unlabeled data

- Can be viewed roughly as a "descriptive statistic", for modular density functions.
- The idea – identify modules in the data, which may be thought of as analogous to unique distributions in a mixture model.
- Used in Exploratory Data Analysis.
- Several clustering algorithms exist (over 100).
- The most appropriate method has to be chosen experimentally.

Introduction

From a Bioinformatics text….

- "Clustering has become so popular in this field most authors present results obtained … and feel the need to include some kind of clustering diagram in their paper."

- The popularity of the clustering techniques is so great that sometimes clustering is mistakenly taken as a very fuzzy and all-inclusive ultimate goal of analysis".

Components of a Clustering Task

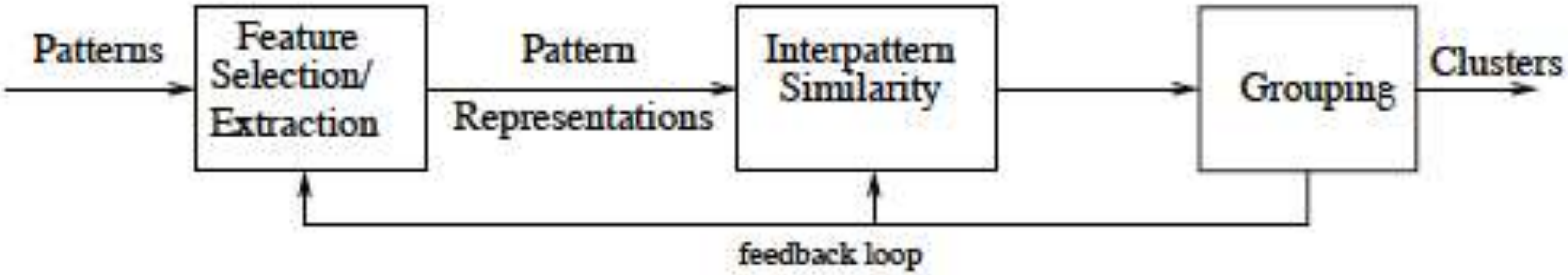


Figure 2. Stages in clustering.

Components of a Clustering Task

1. Pattern Representation (optionally including feature selection).
2. Definition of pattern proximity measure appropriate to the data domain.
3. Clustering or grouping/hierarchical arrangement
4. Data abstraction
5. Assessment of the output (not well defined)

*Humans can detect clusters in two dimensions, but most real problems involve clustering in higher dimensions, where the interpretation of data in the high-dimensional space is more difficult.

*Critical to have understanding of the particular technique being used, and sufficient domain expertise.

Notion of similarity or lack thereof

- Central Metric known as “Similarity” or “Dissimilarity” – measure of likeness between the individual objects being formed. Often drives the algorithms which identify the clusters.

- Similar in spirit to a loss or cost function for the supervised learning task.

Cluster formation

Algorithms proceed in a "bottom-up" or "top-down" fashion.

Agglomerative: (bottom-up) each observation starts in its own cluster, and is merged as one moves up the hierarchy.

Divisive: All observations start in one cluster, and splits are performed divisively as one moves down the cluster (think generalized association rules).

Proximity Measures

- Proximity measures are “Similarity” or “Dissimilarity” matrix $D \in R^{N \times N}$.

- Most algorithms presume a matrix of dissimilarity, with zero diagonal entries, and symmetric.

Dissimilarities

- We have measurements x_{ij} for $i = 1, 2, \ldots, N$, on variables $j = 1, 2, \ldots, p$ (also called attributes).
- First step: construct pairwise dissimilarity between observations, this can be measured in a number of ways.
- We write the dissimilarity $d_j\left(x_{ij}, x_{i'j}\right)$ between values of the j^{th} attribute, and define

$$D\left(x_i, x_{i'}\right) = \sum_{j=1}^{p} d_j\left(x_{ij}, x_{i'j}\right)$$

as the dissimilarity between objects i and i'.
- Most popular squared Euclidean Distance $d\left(x_{ij}, x_{i'j}\right) = \left(x_{ij} - x_{i'j}\right)^2$.

Dissimilarities

- Choice in dissimilarity will lead to different results.
- **Quantitative variables**: squared Euclidean distance, absolute difference, correlation.
- **Ordinal values**: example: grades A, B, C, D, F. Generally defined by replacing their M original values with

$$\frac{i - i/2}{M}, i = 1, \ldots, M$$

  in the prescribed order of the original values. Then they are treated as quantitative [0, 0.5].
- **Categorical variables**: unordered categorical, the degree of difference between pairs of values must be specified, similar to specifying a loss matrix.

Object Dissimilarity

- **Objective**: Combine p-individual attribute dissimilarities $d_j\left(x_{ij}, x_{i'j}\right), j = 1, 2, \ldots, p$ into a single overall measure of dissimilarity $D\left(x_i, x_i'\right)$, produced by a weighted average:

$$D\left(x_i, x_i'\right) = \sum_{j=1}^{p} w_j \cdot d_j\left(x_{ij}, x_{i'j}\right); \quad \sum_{j=1}^{p} w_j = 1.$$

Note: Assigning uniform weights does not necessarily give attributes equal weight, the default weights are relative in terms of the overall contribution.

Object Dissimilarity

- The influence of the *j*th attribute X_j on object dissimilarity $D(x_i, x_{i'})$ depends on its relative contribution to the average object dissimilarity measure over all pairs of observations in the dataset:

$$\bar{D} = \frac{1}{N^2} \sum_{i=1}^{N} \sum_{i'=1}^{N} D(x_i, x_{i'}) = \sum_{j=1}^{p} \boxed{w_j \cdot \bar{d}_j,}$$

Relative influence of the Jth variable is $w_j \cdot \bar{d}_j$ assigning $w_j \sim 1/\bar{d}_j$ would assign equal weights.

where

$$\bar{d}_j = \frac{1}{N^2} \sum_{i=1}^{N} \sum_{i'=1}^{N} d_j(x_{ij}, x_{i'j})$$

is the average dissimilarity on the *j*th attribute.

Object Dissimilarity

Example: For p quantitative variables and a squared Euclidean distance metric:

$$D(x_i, x_{i'}) = \sum_{j=1}^{p} w_j \cdot \boxed{(x_{ij} - x_{i'j})^2}$$

between pairs of points in $\mathbb{R}^p$.

The average dissimilarity, in this case, is given as:

$$\bar{d}_j = \frac{1}{N^2} \sum_{i=1}^{N} \sum_{i'=1}^{N} (x_{ij} - x_{i'j})^2 = 2 \cdot \mathrm{var}_j$$

where var_j is the sample estimate of $Var(X_j)$.

Relative importance is proportional to its relative variance over the data set

Object Dissimilarity

Intuition: Equal weights appear to be conservative.

Reality: It is counterproductive, and ignoring signal in the data.

Variables that are more relevant in separating the groups should be assigned a higher influence in defining object dissimilarity. Assigning equal weights may obscure natural groupings.

**Specifying an appropriate dissimilarity is more important than the algorithm selected for clustering. This selection of dissimilarity pertains more to the domain knowledge.*

Distance Metric

A distance metric, d, is a function that takes as arguments two points x and y, in an n-dimensional space, and has the following properties:

- Symmetry: The distance is symmetric, i.e.,

$$d(x,y) = d(y,x).$$

  This means that the distance from x to y should be the same as the distance from y to x.

- Positivity: The distance between any two points should be a real number greater than or equal to zero:

$$d(x,y) \geq 0$$

  for any x and y. The equality is true iff x=y.

- Triangle Inequality: The distance between two points x and y should be shorter than or equal to the sum of the distances from x to a third point z to y:

$$d(x,y) \leq d(x,z) + d(z,y).$$

Euclidean Distance

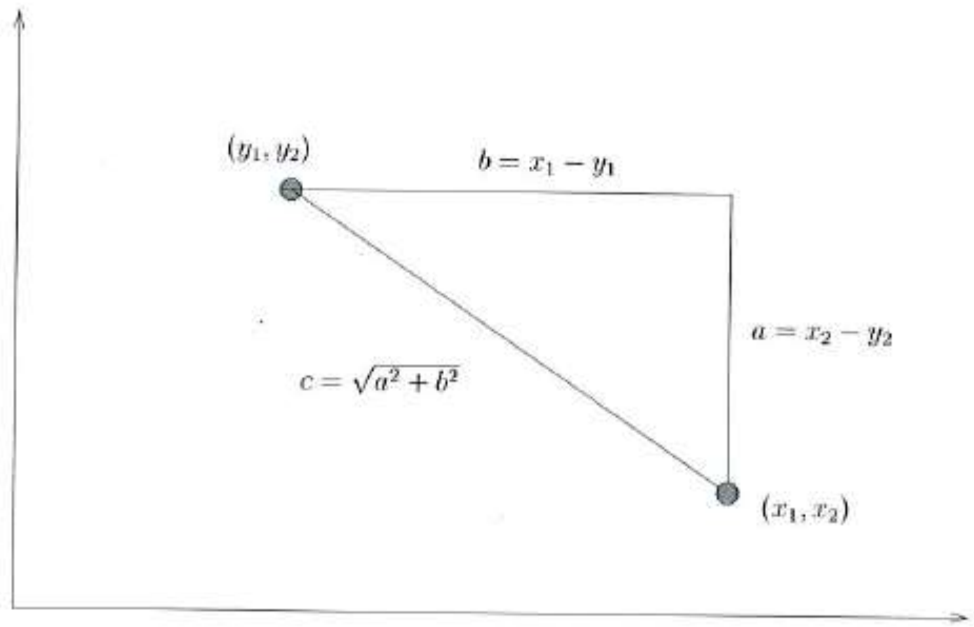
The Euclidean distance between two n-dimensional vectors:

$x = (x_1, x_2, \ldots, x_n)$ and $y = (y_1, y_2, \ldots, y_n)$ is:

$$d(x,y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \cdots (x_N - y_N)^2} = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

From the

Pythagorean theorem



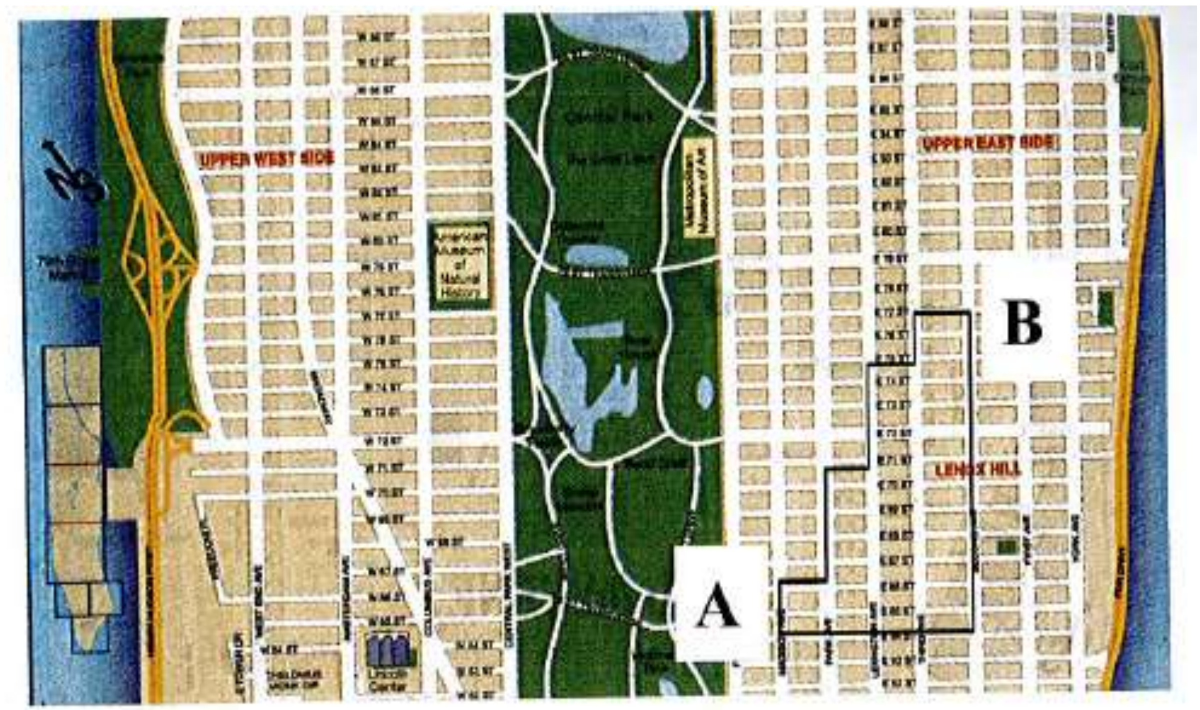
Manhattan Distance (City Block Distance)

The Manhattan distance between two n-dimensional vectors:

$x = (x_1, x_2, \cdots, x_n)$ and $y = (y_1, y_2, \cdots, y_n)$ is:

$$d(x,y) = |x_1 - y_1|^2 + |x_2 - y_2|^2 + \cdots |x_N - y_N|^2 = \sum_{i=1}^{n} |x_i - y_i|^2$$

Can move only along directions Parrallel to x and y. (No diagonal movements)



Manhattan and Euclidean

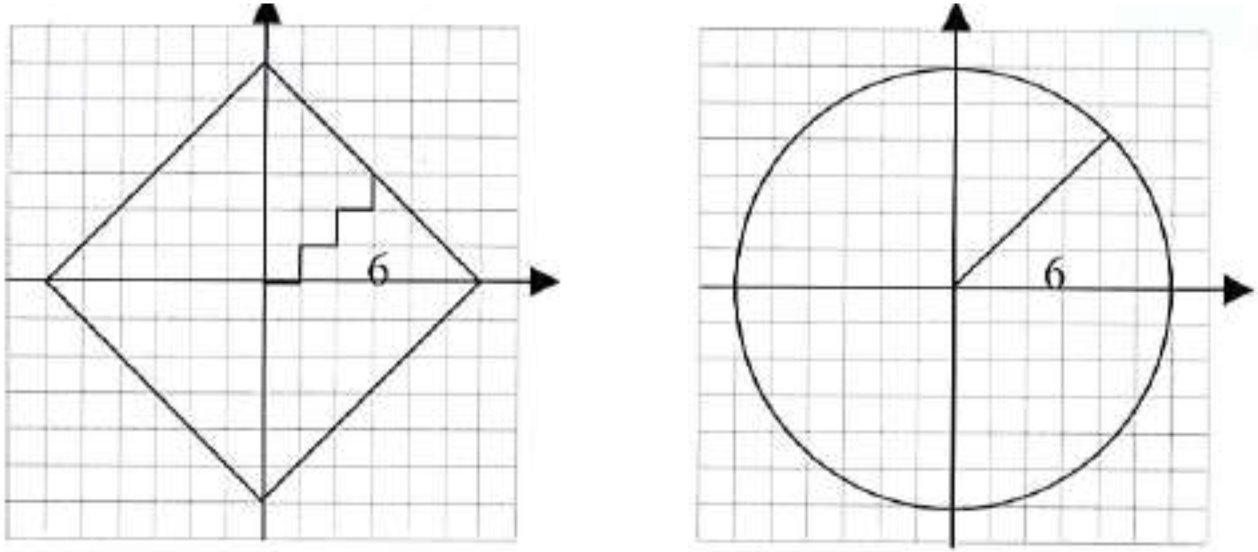


FIGURE 18.4: The distance confers essential properties to the space and objects therein. For instance, a circle is the locus of all points situated at a constant distance from a fixed point called the center. The right panel shows a circle of radius 6 in a Euclidean metric space. The left panel shows the same circle of radius 6 in a Manhattan metric space.

Chebychev Distance

The Chebychev distance between two n-dimensional vectors:

$x = (x_1, x_2, \ldots, x_n)$ and $y = (y_1, y_2, \ldots, y_n)$ is:

$$d(x, y) = \max_i |x_i - y_i|.$$

This is simply the largest difference between any two corresponding coordinates.

Viewed as robust to noise and outliers, as long as the maximum distance is not changed.

Correlation Distance

The Pearson correlation distance between two n-dimensional vectors:

$x = (x_1, x_2, \ldots, x_n)$ and $y = (y_1, y_2, \ldots, y_n)$ is:

$$d(x, y) = 1 - r_{xy}.$$

where
$$r_{xy} = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n} (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^{n} (y_i - \bar{y})^2}}.$$

- Focus' on whether coordinates of points change in the same way.
- Since r is between -1 and 1, the distance is between 0 and 2.
- Sensitive to outlier coordinates (jackknife correlation a work around).

Standardized Euclidean Distance

Standardized Euclidean distance between two n-dimensional vectors:

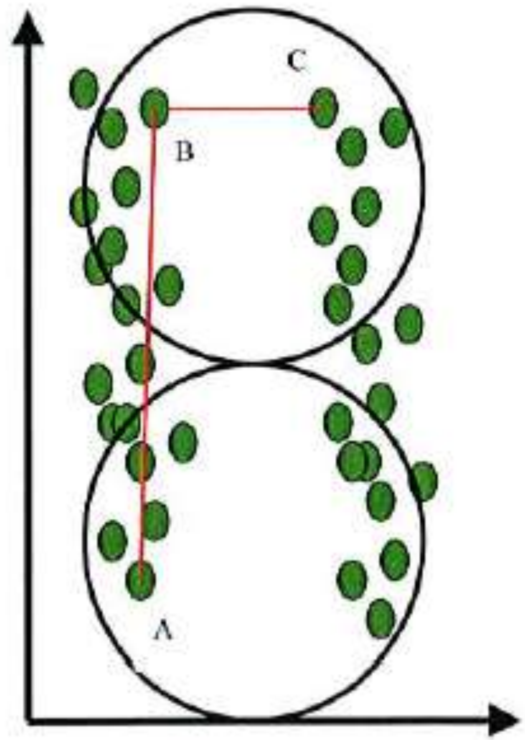
$x = (x_1, x_2, \ldots, x_n)$ and $y = (y_1, y_2, \ldots, y_n)$ is:

$$d(x,y) = \sqrt{\frac{1}{s_1^2}(x_1 - y_1)^2 + \frac{1}{s_2^2}(x_2 - y_2)^2 + \cdots + \frac{1}{s_N^2}(x_N - y_N)^2} = \sqrt{\sum_{i=1}^{N} \frac{1}{s_i^2}(x_i - y_i)^2}$$

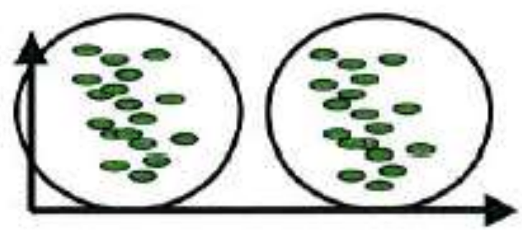
- Not all directions are treated equally.
- Accounts for differences in variances between coordinates.
- Equivalent to distorting the space by shrinking it along the axes of large variance and expanding it along the axes of low variance.

Standardized Euclidean Distance

Euclidean Distance Groups



Standardized
Euclidean Distance Groups



Different variances in coordinate dimensions give rise to the wrong clusters.
Even though A and B are clearly in the same cluster, they are farther apart than B and C.

Mahalanobis Distance

The Mahalanobis distance between two n-dimensional vectors:

$x = (x_1, x_2, \ldots, x_n)$ and $y = (y_1, y_2, \ldots, y_n)$ is:

$$d(x, y) = \sqrt{(x - y)^T S^{-1} (x - y)}$$

Where S is an positive definite matrix.

- Recall: Standardized Euclidean Distance - Equivalent to distorting the space by shrinking it along the aces of large variance and expanding it along the axes of low variance.
- Mahalanobis Distance is more general and achieved through S.

  Often taken to be the sample covariance.

Minkowski Distance

The Minkowski distance between two n-dimensional vectors:

$x = (x_1, x_2, \ldots, x_n)$ and $y = (y_1, y_2, \ldots, y_n)$ is:

$$d(x,y) = \left(|x_1 - y_1|^m + |x_2 - y_2|^m + \cdots + |x_N - y_N|^m\right)^{1/m} = \left(\sum_{i=1}^{N} |x_i - y_i|^m\right)^{1/m}$$

Where S is an positive definite matrix.

- Generalization of the Euclidean (m=2) and Manhattan distance (m=1).

Dissimilarities in Matlab

```
D = pdist(X, DISTANCE) computes D using DISTANCE.  Choices are:

    'euclidean'   - Euclidean distance (default)
    'seuclidean'  - Standardized Euclidean distance. Each coordinate
                    difference between rows in X is scaled by dividing
                    by the corresponding element of the standard
                    deviation S=NANSTD(X). To specify another value for
                    S, use D=pdist(X,'seuclidean',S).
    'cityblock'   - City Block distance
    'minkowski'   - Minkowski distance. The default exponent is 2. To
                    specify a different exponent, use
                    D = pdist(X,'minkowski',P), where the exponent P is
                    a scalar positive value.
    'chebychev'   - Chebychev distance (maximum coordinate difference)
    'mahalanobis' - Mahalanobis distance, using the sample covariance
                    of X as computed by NANCOV. To compute the distance
                    with a different covariance, use
                    D =  pdist(X,'mahalanobis',C), where the matrix C
                    is symmetric and positive definite.
    'cosine'      - One minus the cosine of the included angle
                    between observations (treated as vectors)
    'correlation' - One minus the sample linear correlation between
                    observations (treated as sequences of values).
    'spearman'    - One minus the sample Spearman's rank correlation
                    between observations (treated as sequences of values)
    'hamming'     - Hamming distance, percentage of coordinates
                    that differ
    'jaccard'     - One minus the Jaccard coefficient, the
                    percentage of nonzero coordinates that differ
    function      - A distance function specified using @, for
                    example @DISTFUN.
```

Dissimilarities in R

dist {stats} R Documentation

Distance Matrix Computation

Description

This function computes and returns the distance matrix computed by using the specified distance measure to compute the distances between the rows of a data matrix.

Usage

```
dist(x, method = "euclidean", diag = FALSE, upper = FALSE, p = 2)

as.dist(m, diag = FALSE, upper = FALSE)
## Default S3 method:
as.dist(m, diag = FALSE, upper = FALSE)

## S3 method for class 'dist'
print(x, diag = NULL, upper = NULL,
      digits = getOption("digits"), justify = "none",
      right = TRUE, ...)

## S3 method for class 'dist'
as.matrix(x, ...)
```

Arguments

`x`	a numeric matrix, data frame or `"dist"` object.
`method`	the distance measure to be used. This must be one of `"euclidean"`, `"maximum"`, `"manhattan"`, `"canberra"`, `"binary"` or `"minkowski"`. Any unambiguous substring can be given.

Clustering Algorithms

Objective: partition the observations into groups aka "clusters", such that the pairwise dissimilarities between those assigned to the same cluster tend to be smaller than those in different clusters.

<u>Three types of algorithms</u>

1. **Combinatorial Algorithms** – work directly on the observed data with no reference to an underlying probability model.

1. **Mixture Modeling** – supposes that the data is an i.i.d. sample from some population described by a probability density function. Mixture model, a density for each of the cluster.

1. **Mode Seeking** – "bump hunting" estimation of modes of the pdf, identifying the closet mode as the appropriate cluster.

Clustering: Combinatorial Algorithms

- Most popular clustering paradigm.
- Assign each observation to a group based on a measure, NOT a probability function.
- Each observation is uniquely labeled, $i \in \{1, 2, \ldots, N\}$.
- A pre-specified number of non-overlapping clusters $K < N$ are specified, each given $k \in \{1, \ldots, K\}$ as a number.

  Many to one mapping: $k = C(i)$, which assigns the *i*th observation to the kth cluster.

  <u>Objective</u>: *Seek the mapping* $C^*(i)$ *, which best assigns the observations to clusters based on dissimilarity. The optimal mapping can often be delineated in terms of the assignment of observation to a specific cluster.*

Clustering: Combinatorial Algorithms

Energy based approach, minimize:

$$W(C) = \frac{1}{2}\sum_{k=1}^{K}\sum_{C(i)=k}\sum_{C(i')=k} d(x_i, x_{i'})$$

Analogous to specifying a loss function, in this case, the goal is to assign close points to the same cluster. Characterizes the extent to which observations within the same cluster tend to be close to one another.

Clustering: Combinatorial Algorithms

- Lets break this down, for a more intuitive interpretation.

  Let:

$$T = \frac{1}{2}\sum_{i=1}^{N}\sum_{i'=1}^{N} d_{ii'} = \frac{1}{2}\sum_{k=1}^{K}\sum_{C(i)=k}\left(\sum_{C(i')=k} d_{ii'} + \sum_{C(i')\neq k} d_{ii'}\right)$$

  or

$$T = W(C) + B(C),$$

  where $d_{ii'} = d\left(x_i, x_{i'}\right).$

$$B(C) = \frac{1}{2}\sum_{k=1}^{K}\sum_{C(i)=k}\sum_{C(i')\neq k} d_{ii'}$$

Total point scatter
Constant | Data

Is the in between-cluster point scatter.
Will be large when observations assigned to different clusters are far apart.

- Thus, minimizing $W(C) = T - B(C)$,is equivalent to maximizing $B(C)$.

Clustering: Combinatorial Algorithms

- Now we have a combinatorial optimization problem:

$$W(C) = T - B(C),$$

 we want to minimize W (maximize $B(C)$) over all possible assignments of N data points to K clusters.

- Not possible to enumerate all possibilities and search over a grid, perhaps in tiny data sets. We need an algorithm, or greedy search, for cluster members.

- Number of distinct assignments: $S(N,K) = \frac{1}{K!}\sum_{k=1}^{K}(-1)^{K-k}\binom{K}{k}k^N$

 10 data points, 4 clusters: 34,105.

 19 data points, 4 clusters: 10^10

Clustering: Combinatorial Algorithms

- Algorithms: common theme – greedy search, looking for optimal clusters in an "iterative sense".

- **Note:** this in a combinatorial optimization problem, not a convex optimization problems. Local minima can and will occur.